Enhanced Pareto Local Search for the Satellite Image Mosaic Selection Problem: Diversity-Guided and Scalarization-Guided Parent Selection Strategies

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Abstract

Pareto Local Search (PLS) is a natural and effective local-search framework for multi-objective combinatorial optimisation, but its performance depends critically on how search effort is distributed over the current approximation set. In the four-objective Satellite Image Mosaic Selection (SIMS) problem, a naive PLS policy may spend excessive time exploring densely populated regions of the archive while neglecting other regions that are either structurally underrepresented or more promising under specific trade-off directions.

We study two archive-steering enhancements to a common improved PLS baseline. The first is a *diversity-guided probing* strategy that selects a well-spread subset of candidate parents in objective space before neighbourhood exploration. The second is a *scalarization-guided selection* strategy that repeatedly samples weighted Chebycheff trade-off directions and explores archive solutions that are best aligned with those directions. Both mechanisms operate at the level of parent-solution scheduling rather than neighbourhood definition, and therefore complement the standard PLS framework without changing the underlying local move operator.

Experiments on 20 benchmark instances (five cities, four size tiers, 30–150 images, four objectives) show that both enhancements improve hypervolume over the baseline on medium and large instances, but that diversity-guided probing is the stronger default strategy overall. Across all 20 instances, diversity-guided probing improves final normalised hypervolume by +3.60% on average over the baseline, whereas scalarization-guided selection improves it by +3.42%. Direct comparison shows that diversity-guided probing outperforms scalarization-guided selection on 18 of 20 instances, with an average advantage of 0.18%. Scalarization-guided selection nevertheless achieves the best result on the hardest Lagos instance (+3.06% over diversity-guided probing at 145 images), indicating that directional archive steering can be particularly effective on certain large, structurally heterogeneous instances.

Keywords: multi-objective optimisation, Pareto local search, satellite image selection, hypervolume, diversity-guided search, scalarization-guided search, weighted Chebycheff

1 Introduction

The *Satellite Image Mosaic Selection* (SIMS) problem requires choosing a subset of satellite images that jointly cover a region of interest while balancing several conflicting criteria. In the four-objective setting considered here, the optimisation simultaneously minimises acquisition cost, cloud contamination, worst-case incidence angle, and resolution loss. The resulting problem is a multi-objective variant of weighted set cover and is computationally challenging even for moderate catalogue sizes.

Pareto Local Search (PLS) is a natural heuristic for this setting. It maintains an archive of mutually non-dominated solutions and repeatedly explores neighbourhoods of selected archive members. The quality of the final approximation set depends not only on the neighbourhood operator itself, but also on the *policy used to decide which archive solutions should be explored next*. In many-objective settings, this policy becomes especially important because the archive can grow rapidly, making exhaustive and uniformly distributed exploration increasingly inefficient.

This paper focuses on two search-allocation enhancements that operate at the parent-selection level:

1. **Diversity-guided probing**, which selects a well-spread subset of archive or population members in objective space before neighbourhood exploration.
2. **Scalarization-guided selection**, which samples weighted trade-off directions and prioritises archive solutions that minimise the corresponding weighted Chebycheff scalarizations.

Both strategies are layered on top of the same improved PLS baseline and therefore isolate the effect of *search steering* rather than neighbourhood design. The goal is not to alter the definition of local moves, but to improve the *allocation of limited search effort* over the current approximation set.

Contributions. This paper makes the following contributions:

1. We formalise two parent-selection strategies for PLS at the same level of abstraction as the baseline algorithm: a diversity-guided strategy and a scalarization-guided strategy.
2. We provide a controlled empirical comparison of these strategies against the same baseline PLS on 20 four-objective SIMS instances.
3. We show that diversity-guided probing is the stronger default strategy overall, while scalarization-guided selection can dominate on selected large instances.
4. We analyse the results by instance size and identify the regimes in which each strategy is most effective.

2 Problem Formulation

We consider a finite set of candidate images $\mathcal{I} = \{1, \ldots, m\}$ and a finite set of ground elements $\mathcal{E} = \{1, \ldots, n\}$ that must be covered. Each image $i \in \mathcal{I}$ covers a subset $C_i \subseteq \mathcal{E}$. A feasible solution is any subset $S \subseteq \mathcal{I}$ such that

$$\bigcup_{i \in S} C_i = \mathcal{E}.$$

2.1 Decision Space

A solution is represented by a binary vector $x \in \{0, 1\}^m$, where $x_i = 1$ if image i is selected and $x_i = 0$ otherwise. The feasible set is

$$\mathcal{F} = \{x \in \{0, 1\}^m \mid \forall e \in \mathcal{E},\ \exists i \in \mathcal{I} \text{ such that } x_i = 1 \text{ and } e \in C_i\}.$$

2.2 Objective Space

Each feasible solution is evaluated by four minimisation objectives:

$$f(x) = \big(f_1(x), f_2(x), f_3(x), f_4(x)\big),$$

where:

- f_1: total acquisition cost,
- f_2: total cloud contamination,
- f_3: worst-case incidence angle,
- f_4: resolution loss.

For two feasible solutions $x, y \in \mathcal{F}$, we say that x *dominates* y, written $x \prec y$, if

$$\forall j \in \{1, \ldots, 4\},\ f_j(x) \le f_j(y) \quad \text{and} \quad \exists j \in \{1, \ldots, 4\},\ f_j(x) < f_j(y).$$

The goal is to approximate the Pareto set of mutually non-dominated feasible solutions.

3 Baseline PLS

PLS maintains two sets:

- an archive $\mathcal{A}$ of mutually non-dominated solutions found so far, and
- a working population $\mathcal{P} \subseteq \mathcal{A}$ of solutions whose neighbourhoods are scheduled for exploration.

At each iteration, the algorithm selects one or more parent solutions from $\mathcal{P}$, explores their neighbourhoods, inserts newly discovered non-dominated neighbours into $\mathcal{A}$, and forms the next working population from the accepted offspring. The baseline policy considered here uses the same neighbourhood operator and archive update mechanism throughout all experiments; only the *parent-selection strategy* changes.

3.1 Neighbourhood Operator

Let $x \in \mathcal{F}$ be a feasible solution. Its neighbourhood $\mathcal{N}(x)$ consists of feasible solutions obtained by local image-set modifications that preserve full coverage. The exact move operator is not the focus of this paper; what matters is that all compared configurations use the same neighbourhood definition and therefore differ only in how parent solutions are chosen for exploration.

3.2 Main Loop

At a high level, baseline PLS can be written as follows.

Algorithm 1 Baseline Pareto Local Search

Input: Initial feasible population $\mathcal{P}_0$, neighbourhood operator $\mathcal{N}$
Output: Approximation archive $\mathcal{A}$

$\mathcal{A} \leftarrow \textsc{NonDominated}(\mathcal{P}_0)$
$\mathcal{P} \leftarrow \mathcal{A}$
while termination criterion not met **do**
    choose parent subset $\mathcal{S} \subseteq \mathcal{P}$
    $\mathcal{P}' \leftarrow \emptyset$
    for each $x \in \mathcal{S}$ **do**
        for each $y \in \mathcal{N}(x)$ **do**
            if y is not dominated by $\mathcal{A}$ **then**
                insert y into $\mathcal{A}$, removing dominated archive members
                insert y into $\mathcal{P}'$
    $\mathcal{P} \leftarrow \textsc{NextPopulation}(\mathcal{P}, \mathcal{P}', \mathcal{A})$
return $\mathcal{A}$

The central question addressed in this paper is therefore: *how should the parent subset $\mathcal{S}$ be chosen?*

4 Enhancement 1: Diversity-Guided Probing

4.1 Motivation

A straightforward parent-selection policy may repeatedly explore archive members that are close to one another in objective space. This can cause search effort to concentrate on already dense regions of the current approximation set while leaving other regions underexplored. When the archive is large, such concentration reduces the effective coverage of the Pareto front approximation.

The purpose of diversity-guided probing is to allocate exploration effort to a subset of parent solutions that is *well spread* in objective space. The underlying principle is that, under a fixed computational budget, it is often preferable to explore a representative set of structurally different archive members rather than many nearly redundant ones.

4.2 Method

Let $\mathcal{P}$ denote the current candidate parent set. Diversity-guided probing selects a subset $\mathcal{S} \subseteq \mathcal{P}$ of prescribed budget B such that the selected solutions are as far apart as possible in normalised objective space.

The method proceeds conceptually in three stages:

1. **Normalisation.** Objective vectors are mapped to a common scale using the current objective ranges in the candidate set.
2. **Seed selection.** One candidate is chosen as an initial representative.
3. **Farthest-point expansion.** Repeatedly add the candidate whose minimum distance to the already selected set is maximal.

This produces a subset that approximates a max–min spread criterion. The selected parents are then explored in the order in which they were chosen, so that even if the run is interrupted early, the explored subset remains broadly representative.

4.3 Theoretical Effect

Diversity-guided probing does not change the archive update rule or the definition of local optimality. Instead, it changes the *order and distribution* of neighbourhood exploration. Its expected effect is:

- broader coverage of the current approximation set,
- reduced oversampling of dense regions,
- improved anytime behaviour under fixed budgets.

The method is especially attractive when the archive is already large and the main challenge is not finding *any* improving move, but deciding *where* to spend the next unit of search effort.

5 Enhancement 2: Scalarization-Guided Parent Selection

5.1 Motivation

Diversity alone does not distinguish between archive members that are merely different and archive members that are particularly promising under a given trade-off direction. In many-objective search, a useful alternative is to steer exploration by repeatedly sampling preference directions and selecting archive solutions that are best aligned with those directions.

The scalarization-guided strategy follows this idea. Rather than choosing parents solely for geometric spread, it samples weighted trade-off vectors and prioritises archive members that minimise the corresponding scalarized objective. This biases the search toward solutions that are likely to improve the approximation set in specific regions of the Pareto front.

5.2 Method

Let $z^\star$ denote a reference ideal point and let $\lambda = (\lambda_1, \ldots, \lambda_D)$ be a non-negative weight vector with $\sum_j \lambda_j = 1$. For a solution x, define the weighted Chebycheff scalarization

$$g_\lambda(x \mid z^\star) = \max_{j=1,\ldots,D} \left\{ \lambda_j \frac{f_j(x) - z_j^\star}{r_j} \right\},$$

where $r_j > 0$ is a normalising range for objective j.

At each parent-selection step:

1. sample one or more weight vectors λ,
2. for each sampled λ, identify the archive solution $x^\lambda \in \mathcal{A}$ minimising $g_\lambda(x \mid z^\star)$,
3. explore the neighbourhoods of the selected minimisers.

This strategy is archive-driven: the parent candidates are drawn from the current non-dominated archive rather than from an arbitrary working population. In this way, the search is repeatedly redirected toward solutions that are most representative of sampled trade-off directions.

5.3 Theoretical Effect

Scalarization-guided selection changes the search policy from *coverage-oriented* to *direction-oriented*. Its expected effect is:

- stronger exploitation of promising trade-off directions,
- improved progress in regions where the archive is sparse but strategically important,
- better alignment between parent selection and quality indicators based on directional trade-offs.

The method is particularly appealing when the archive is large and the main challenge is to identify which archive members are most likely to yield useful improvements under limited exploration budgets.

6 Experimental Setup

6.1 Instances

We use 20 benchmark instances derived from real satellite catalogues over five cities (Lagos, Mexico City, Paris, Rio de Janeiro, and Tokyo Bay) at four size tiers up to 150 images. Lagos uses 145 images at the largest tier, while the other cities use 150 images. All instances are optimised with $D = 4$ objectives.

Table 1: *Benchmark instance characteristics.*

City	Images	Elements	Timeout (s)
Lagos	30	443	300
	50	1 147	300
	100	4 420	1200
	145	8 503	1200
Mexico City	30	333	300
	50	842	300
	100	3 384	1200
	150	5 236	1200
Paris	30	475	300
	50	970	300
	100	4 503	1200
	150	11 027	1200
Rio de Janeiro	30	351	300
	50	1 238	300
	100	5 905	1200
	150	12 879	1200
Tokyo Bay	30	298	300
	50	806	300
	100	3 278	1200
	150	8 079	1200

6.2 Configurations

We compare three configurations:

1. **Pure PLS:** the baseline PLS reference.
2. **Diverse Probe PLS:** the same baseline augmented with diversity-guided parent probing.
3. **Scalarized PLS:** the same baseline augmented with scalarization-guided archive selection.

The purpose of this design is to isolate the effect of the two parent-selection strategies while keeping the underlying local search framework fixed.

6.3 Metrics

The primary metric is the **normalised hypervolume** (HV) computed with shared per-instance bounds derived from the union of all trace points across the compared configurations. This ensures that HV values are directly comparable.

We also analyse **HV-over-time** curves reconstructed from the search traces. For each configuration and instance, the trace records the sequence of non-dominated-at-discovery solutions together with their timestamps. We reconstruct the Pareto front at evenly spaced time points and compute HV at each point.

6.4 Implementation

The experiment driver is implemented in Python and calls a compiled PLS solver through language bindings. Hypervolume-over-time curves are computed from archived traces using a shared reconstruction and evaluation pipeline. All compared configurations use the same implementation framework and differ only in the parent-selection policy.

7 Results

7.1 Hypervolume Summary

Table 2 reports the final normalised hypervolume for all 20 instances. The table includes the baseline, the diversity-guided variant, and the scalarization-guided variant.

Several patterns emerge immediately:

- Both enhanced strategies substantially outperform the baseline on many medium and large instances.
- Diversity-guided probing is the strongest overall strategy, achieving the best HV on 18 of 20 instances.
- Scalarization-guided selection nevertheless achieves the best result on two large instances: Lagos 145 and Rio 150.
- On the smallest instances, all three configurations are often very close, indicating that the baseline already explores the search space effectively under the available budget.

7.2 Improvement over the Baseline

To quantify the gains over the baseline, Table 3 reports the percentage improvement of each enhanced strategy relative to Pure PLS.

The average improvements show that both strategies are beneficial, but that diversity-guided probing has a slight overall advantage. The difference is small in aggregate, yet consistent enough to make diversity-guided probing the stronger default choice.

7.3 Direct Comparison: Diverse vs Scalarized

Table 4 compares the two enhanced strategies directly.

The direct comparison is decisive:

- Diversity-guided probing wins on 18 of 20 instances.
- Scalarization-guided selection wins on only two instances, both at the largest size tier.
- The average deficit of scalarization-guided selection relative to diversity-guided probing is 0.18%.

Thus, while scalarization-guided selection is competitive and sometimes superior on difficult large instances, diversity-guided probing is the better general-purpose strategy on this benchmark set.

Table 2: *Final normalised hypervolume for Pure PLS, Diverse Probe PLS, and Scalarized PLS. Positive Δ values indicate improvement over the Pure PLS baseline.*

Instance	**Size**	**Pure PLS**	**Diverse Probe**	**Scalarized**	**Best**
Lagos 30	30	0.811595	0.811595	0.811576	Diverse
Lagos 50	50	0.868126	0.867881	0.867573	Pure
Lagos 100	100	0.886549	0.923981	0.922884	Diverse
Lagos 145	145	0.734004	0.792120	0.816380	Scalarized
Mexico City 30	30	0.942413	0.942413	0.942400	Diverse
Mexico City 50	50	0.923096	0.922876	0.921505	Pure
Mexico City 100	100	0.902607	0.924284	0.921899	Diverse
Mexico City 150	150	0.784194	0.883515	0.879431	Diverse
Paris 30	30	0.839079	0.838858	0.821789	Pure
Paris 50	50	0.853323	0.855010	0.852591	Diverse
Paris 100	100	0.782960	0.894449	0.880452	Diverse
Paris 150	150	0.875556	0.944785	0.943737	Diverse
Rio 30	30	0.812803	0.812803	0.812541	Diverse
Rio 50	50	0.643427	0.643056	0.642872	Pure
Rio 100	100	0.936971	0.938717	0.937587	Diverse
Rio 150	150	0.781738	0.876039	0.876232	Scalarized
Tokyo Bay 30	30	0.719423	0.719423	0.719303	Diverse
Tokyo Bay 50	50	0.848188	0.848105	0.839300	Pure
Tokyo Bay 100	100	0.880687	0.898726	0.897927	Diverse
Tokyo Bay 150	150	0.860256	0.932349	0.928658	Diverse

Table 3: *Percentage hypervolume improvement over Pure PLS.*

Instance	Size	Diverse vs Pure (%)	Scalarized vs Pure (%)
Lagos 30	30	+0.00	-0.00
Lagos 50	50	-0.03	-0.06
Lagos 100	100	+4.22	+4.10
Lagos 145	145	+7.92	+11.22
Mexico City 30	30	+0.00	-0.00
Mexico City 50	50	-0.02	-0.17
Mexico City 100	100	+2.40	+2.14
Mexico City 150	150	+12.67	+12.14
Paris 30	30	-0.03	-2.06
Paris 50	50	+0.20	-0.09
Paris 100	100	+14.24	+12.45
Paris 150	150	+7.91	+7.79
Rio 30	30	+0.00	-0.03
Rio 50	50	-0.06	-0.09
Rio 100	100	+0.19	+0.07
Rio 150	150	+12.06	+12.09
Tokyo Bay 30	30	+0.00	-0.02
Tokyo Bay 50	50	-0.01	-1.05
Tokyo Bay 100	100	+2.05	+1.96
Tokyo Bay 150	150	+8.38	+7.95
Average		**+3.60**	**+3.42**

Table 4: *Direct comparison of Scalarized PLS against Diverse Probe PLS. Positive Δ indicates that Scalarized PLS is better.*

Instance	**Size**	**Scalarized – Diverse (%)**	**Winner**
Lagos 30	30	-0.00	Diverse
Lagos 50	50	-0.04	Diverse
Lagos 100	100	-0.12	Diverse
Lagos 145	145	+3.06	Scalarized
Mexico City 30	30	-0.00	Diverse
Mexico City 50	50	-0.15	Diverse
Mexico City 100	100	-0.26	Diverse
Mexico City 150	150	-0.46	Diverse
Paris 30	30	-2.03	Diverse
Paris 50	50	-0.28	Diverse
Paris 100	100	-1.56	Diverse
Paris 150	150	-0.11	Diverse
Rio 30	30	-0.03	Diverse
Rio 50	50	-0.03	Diverse
Rio 100	100	-0.12	Diverse
Rio 150	150	+0.02	Scalarized
Tokyo Bay 30	30	-0.02	Diverse
Tokyo Bay 50	50	-1.04	Diverse
Tokyo Bay 100	100	-0.09	Diverse
Tokyo Bay 150	150	-0.40	Diverse
Average		**-0.18**	**Diverse**
Win–loss		2 wins, 18 losses	

7.4 Hypervolume-Over-Time Analysis

The hypervolume-over-time curves confirm the same qualitative picture. The diversity-guided strategy tends to achieve stronger early coverage of the approximation set, while the scalarization-guided strategy is more selective and therefore sometimes slower to spread across the front. On the hardest instances, however, scalarization-guided selection can eventually overtake diversity-guided probing by focusing search effort on strategically important trade-off directions.

7.5 Visual Summary

Figure 2 summarises the final HV values across all instances.

8 Discussion

Why does diversity-guided probing work so well? The SIMS objective space is heterogeneous and often exhibits multiple structurally distinct trade-off regions. A diversity-guided policy allocates search effort across these regions more evenly, which is especially valuable when the archive is already large and the main risk is oversampling dense regions. This explains why diversity-guided probing is the stronger default strategy on most instances.

When does scalarization-guided selection help? Scalarization-guided selection is most effective when the search must be steered toward specific trade-off directions that are difficult to reach through broad coverage alone. The Lagos 145 and Rio 150 instances illustrate this regime: on these large instances, directional archive steering yields the best final HV.

Small-instance behaviour. On the 30-image tier, the differences between the strategies are minimal. This suggests that the baseline already explores the search space sufficiently well under the available budget, leaving little room for parent-selection improvements.

Medium and large instances. From 100 images onward, both enhancements become clearly beneficial over the baseline. The gains are particularly strong on Paris 100, Mexico City 150, Rio 150, and Tokyo Bay 150, indicating that parent-selection policy matters most when the archive is large and the search budget must be allocated carefully.

Practical recommendation. If a single enhanced strategy must be chosen, the results support *diversity-guided probing* as the default. Scalarization-guided selection remains valuable as a complementary strategy, especially for large instances where directional exploitation can dominate broad coverage.

9 Related Work

Pareto Local Search has long been recognised as a strong local-search framework for multi-objective combinatorial optimisation. A central theme in the literature is that PLS performance depends not only on the neighbourhood operator, but also on archive management and search scheduling.

Diversity-oriented archive exploration is closely related to the broader idea of maintaining spread in multi-objective search, which appears in evolutionary algorithms, decomposition methods, and archive management schemes. The diversity-guided probing strategy studied here adapts this principle to the parent-selection stage of PLS.

Scalarization-guided search has an equally rich history. Weighted Chebycheff and related scalarizations are standard tools for expressing trade-off directions in multi-objective optimisation. In the present context, scalarization is not used to replace Pareto dominance, but to guide the order in which archive members are selected for local exploration.

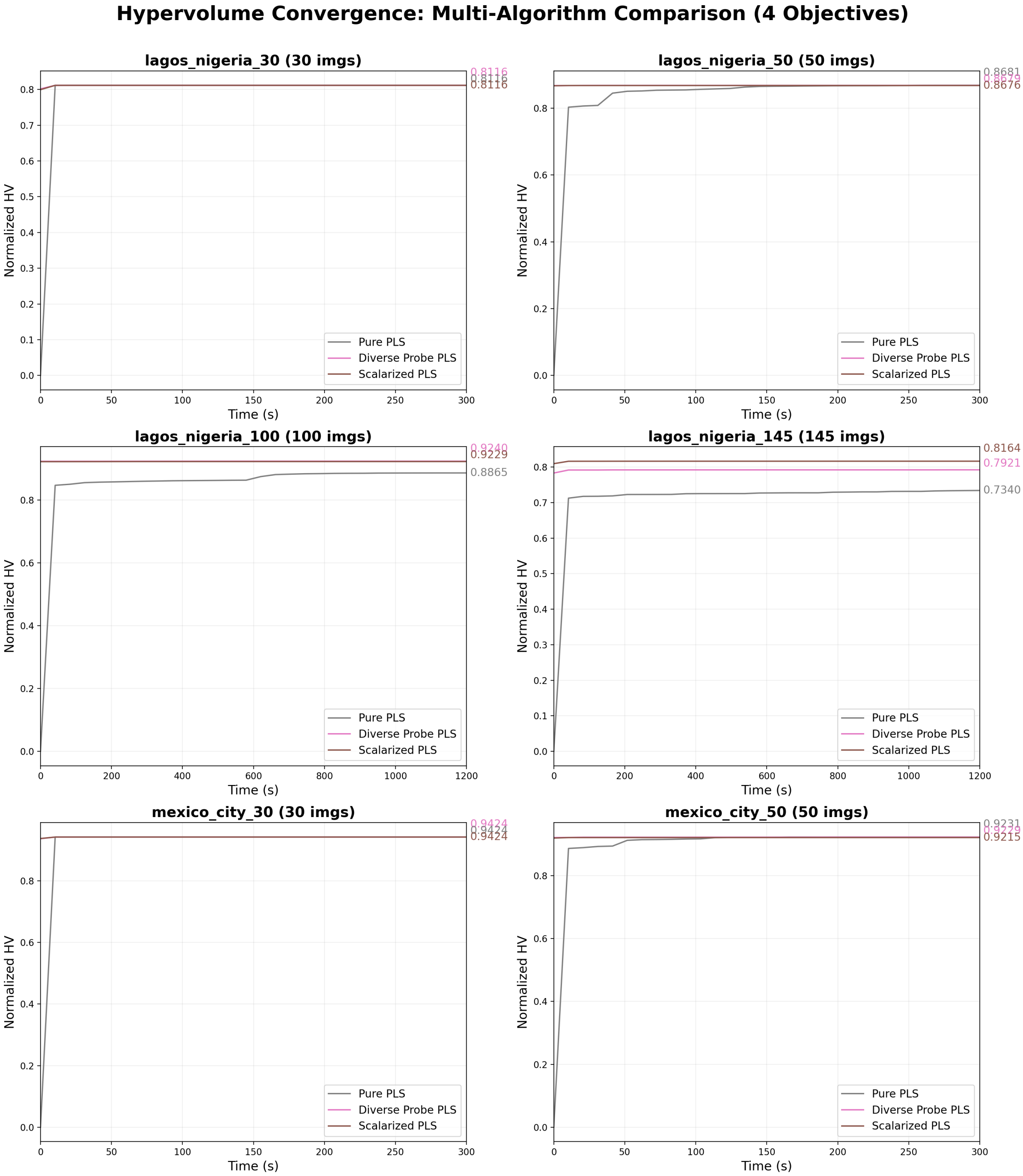

Figure 1: *Normalised hypervolume over time for all benchmark instances in the Diverse Probe PLS vs Scalarized PLS comparison.*

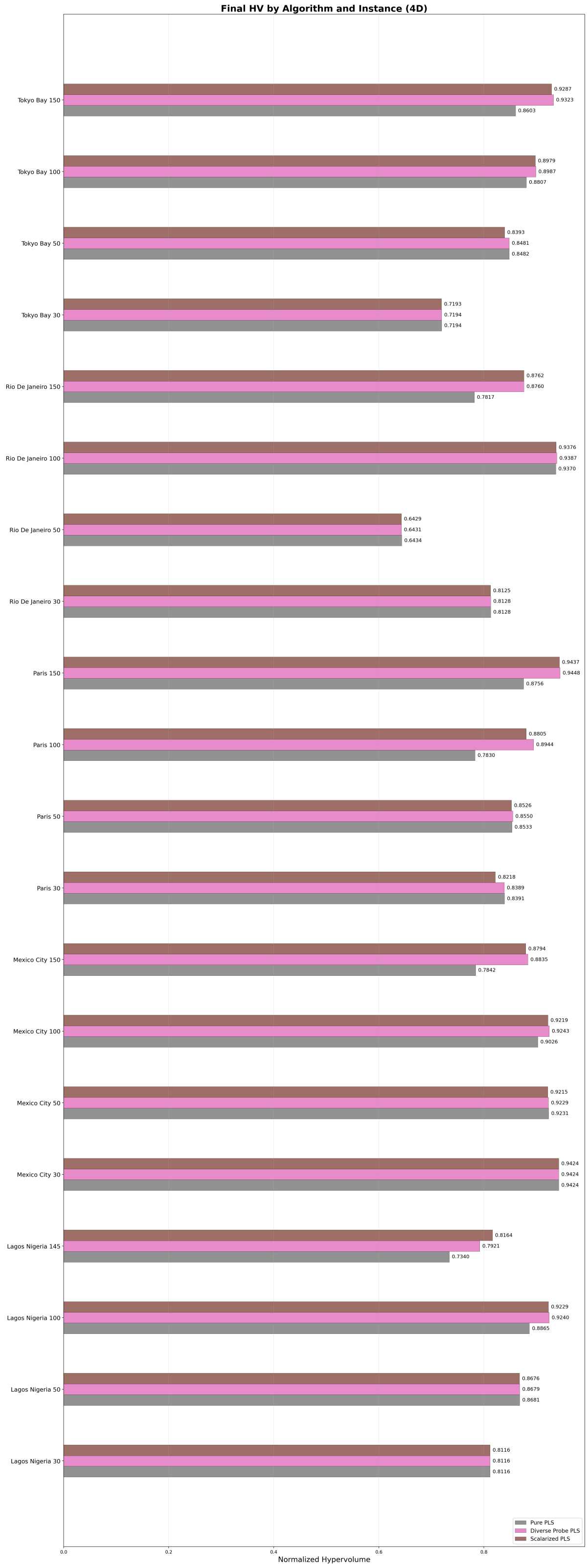

Figure 2: *Final normalised hypervolume by instance for Pure PLS, Diverse Probe PLS, and Scalarized PLS.*

The present study therefore sits at the intersection of two established ideas—diversity preservation and scalarization-based guidance—and evaluates them in a controlled PLS setting for the SIMS problem.

10 Conclusion

We presented a comparative study of two parent-selection enhancements for Pareto Local Search on the four-objective Satellite Image Mosaic Selection problem: diversity-guided probing and scalarization-guided archive selection.

The experiments show that:

- both strategies improve over the baseline on medium and large instances,
- diversity-guided probing is the stronger default strategy, improving final HV by +3.60% on average over the baseline,
- scalarization-guided selection is also effective, improving final HV by +3.42% on average over the baseline,
- direct comparison favours diversity-guided probing on 18 of 20 instances, although scalarization-guided selection wins on two large instances.

These results suggest that broad archive coverage is generally more valuable than directional exploitation in this benchmark set, but that scalarization-guided steering remains an important complementary tool for difficult large instances.

Future work. Natural next steps include:

1. adaptive switching between diversity-guided and scalarization-guided parent selection during a run,
2. hybrid parent-selection policies that combine spread and directional preference more tightly,
3. evaluation on the 200-image instances and under larger time budgets,
4. extension of the same comparison to hybrid exact/heuristic configurations.

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